

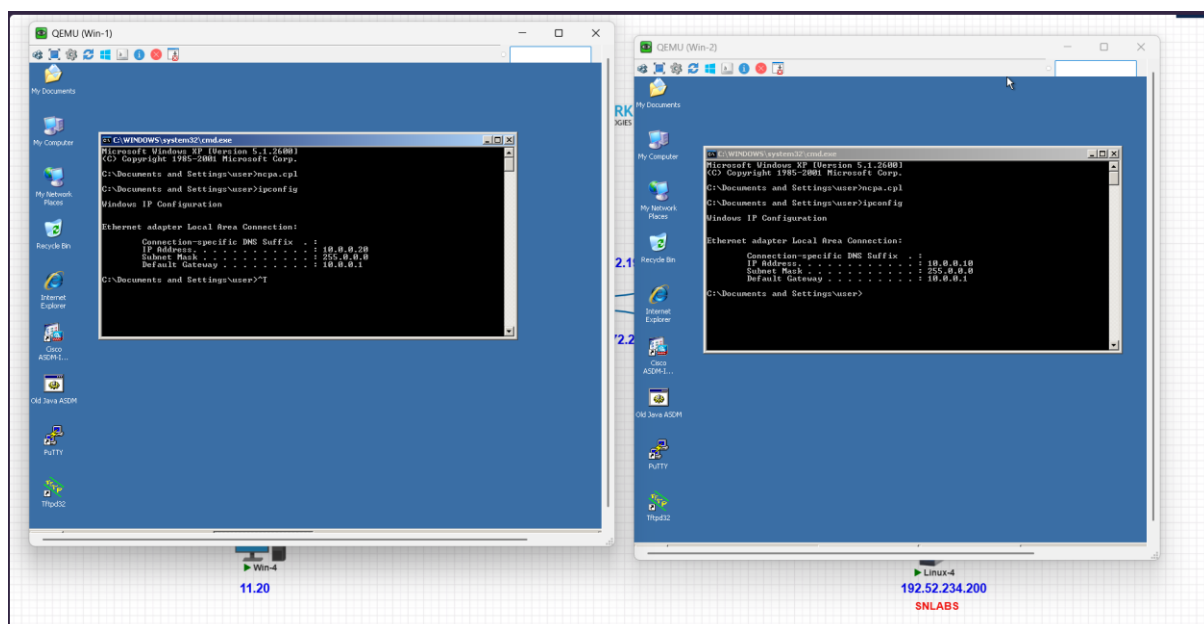
ACL PROJECT

- Scenario 1 :** Allow 10.0.0.0 to communicate only instagram server using port 80
- Scenario 2 :** Allow ping communication from 10.0.0.10 to instagram server
- Scenario 3 :** Allow 10.0.0.20 to communicate facebook server using ssh & http
- Scenario 4 :** Allow 10.0.0.0 to communicate snsite server using http
- Scenario 5 :** Allow 10.0.0.10 to communicate snlabs server using http & icmp
- Scenario 6 :** Allow 11.0.0.0 to communicate instagram & facebook server using http
- Scenario 7 :** Allow 11.0.0.10 to ping facebook server
- Scenario 8 :** Allow 11.0.0.10 to ping snlab server
- Scenario 9 :** Allow 11.0.0.0 to access snlab server
- Scenario 10 :** Deny 11.0.0.0 to access 10.0.0.0

TASK 1: ASSIGNING IP ADDRESS TO PCs

PC1 IP: 10.0.0.20

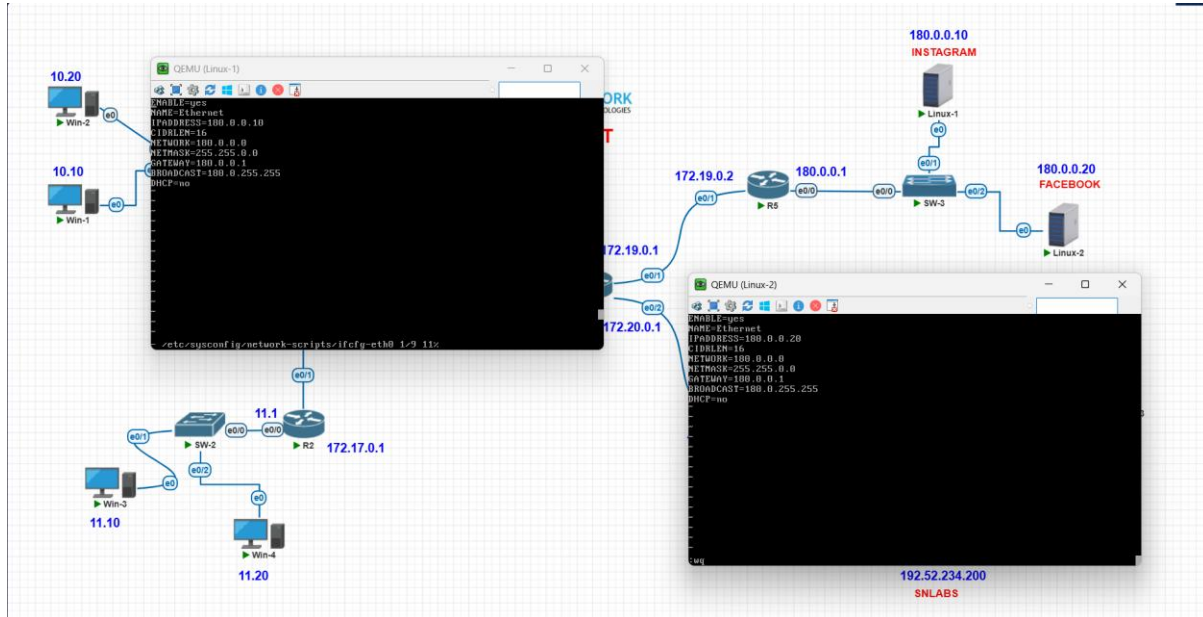
PC2 IP: 10.0.0.10



TASK 2: ASSIGNING IP ADDRESS TO LINUX SERVERS

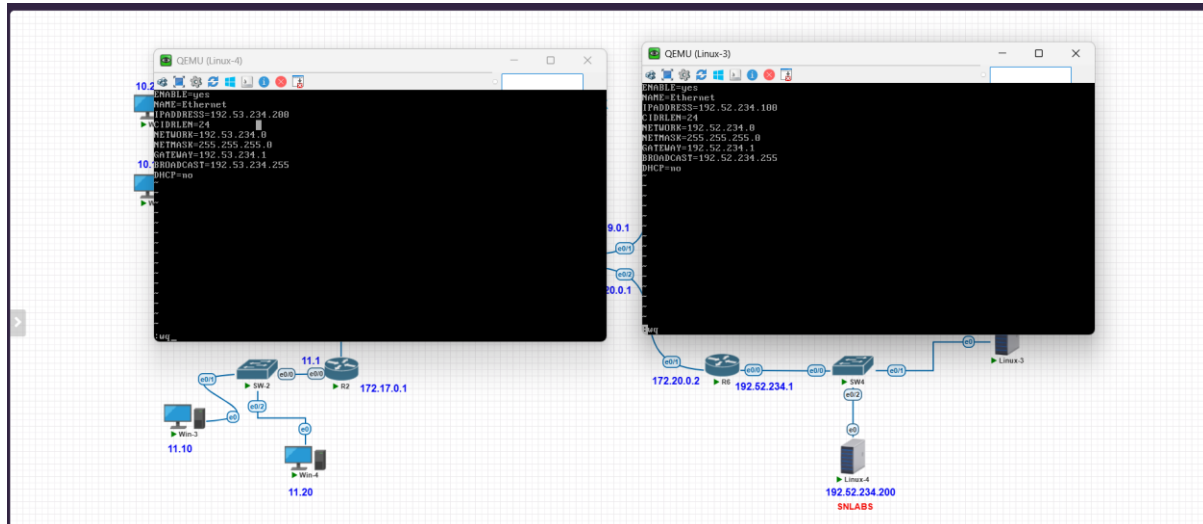
SERVER1 IP: 180.0.0.10

SERVER2 IP: 180.0.0.20

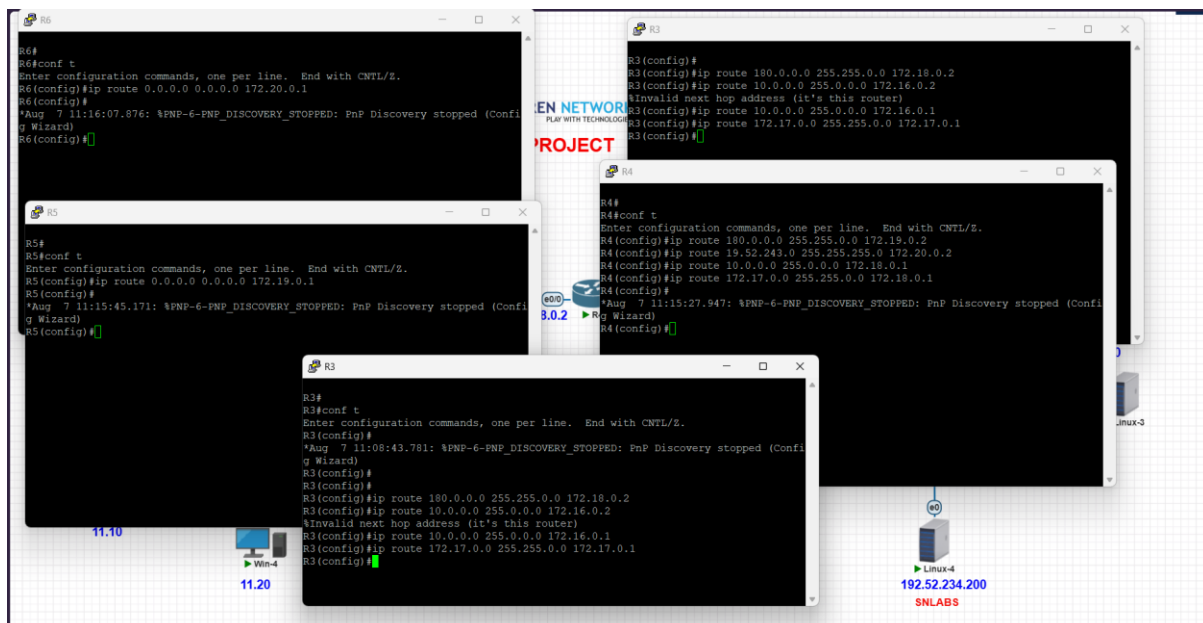


SERVER3 IP: 192.52.234.100

SERVER4 IP: 192.52.234.200



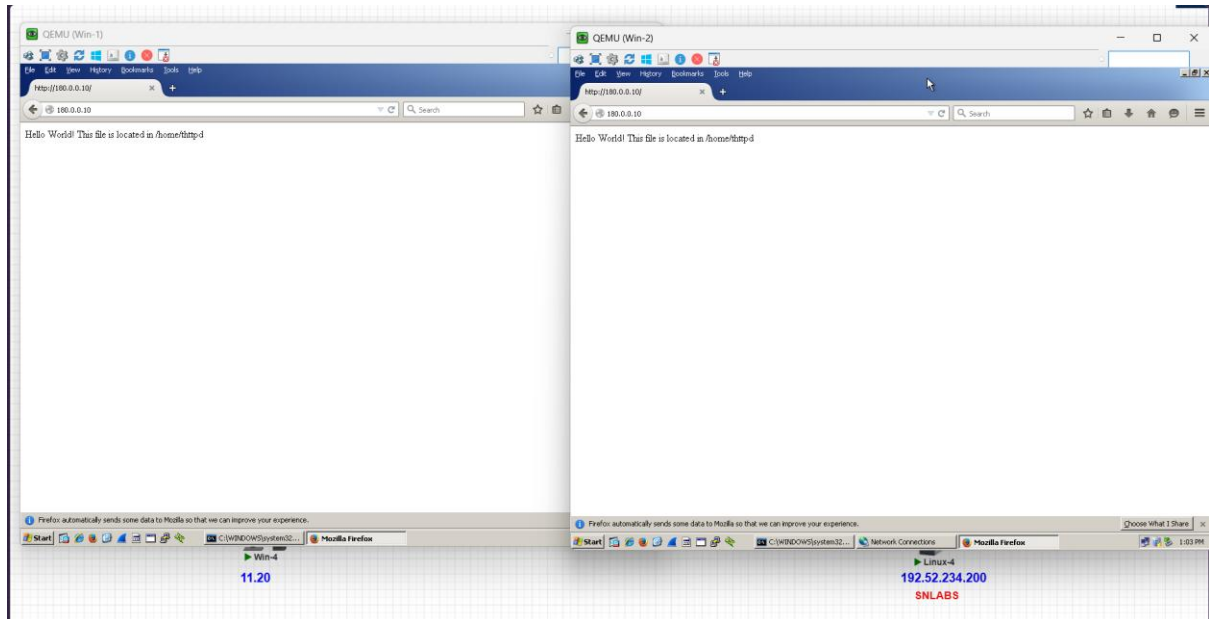
TASK 3: ADDING DEFAULT ROUTES (0.0.0.0/0) IN ALL THE ROUTERS



TASK 5: ACL CONFIGURATION IN ROUTER 1 BASED ON SCENARIO

SCENARIO 1

Allow 10.0.0.0 to communicate only instagram server using port 80

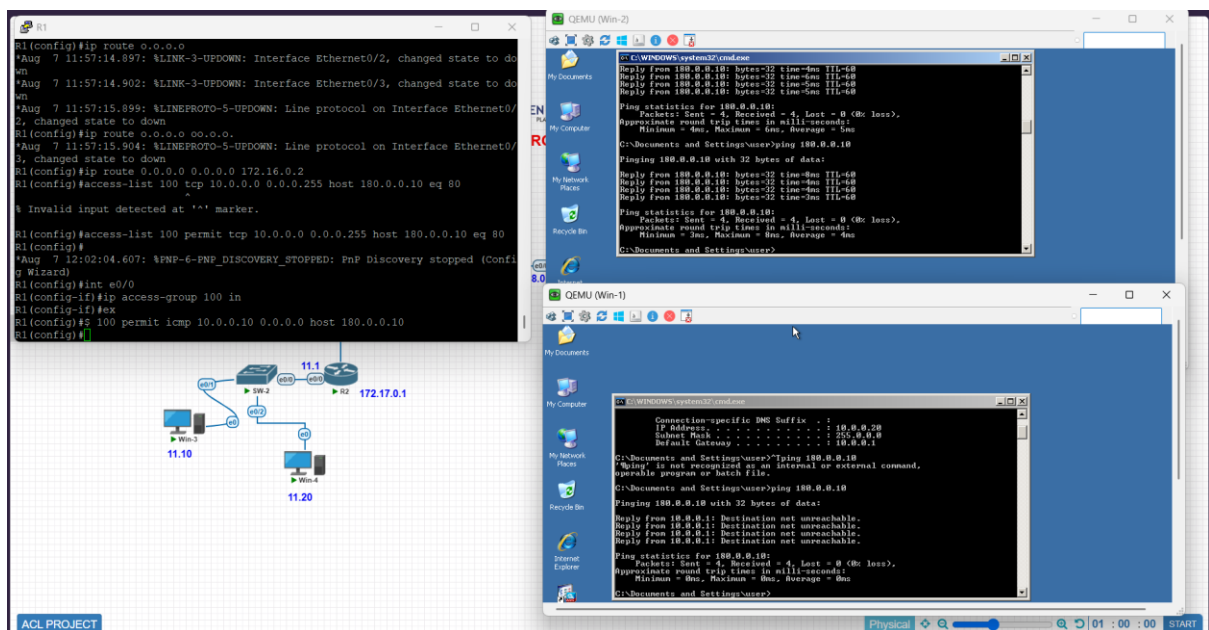


SCENARIO 2

Allow ping communication from 10.0.0.10 to instagram server

ACL CONFIGURATION IN ROUTER

RESPONSES FROM PC 1 & PC 2

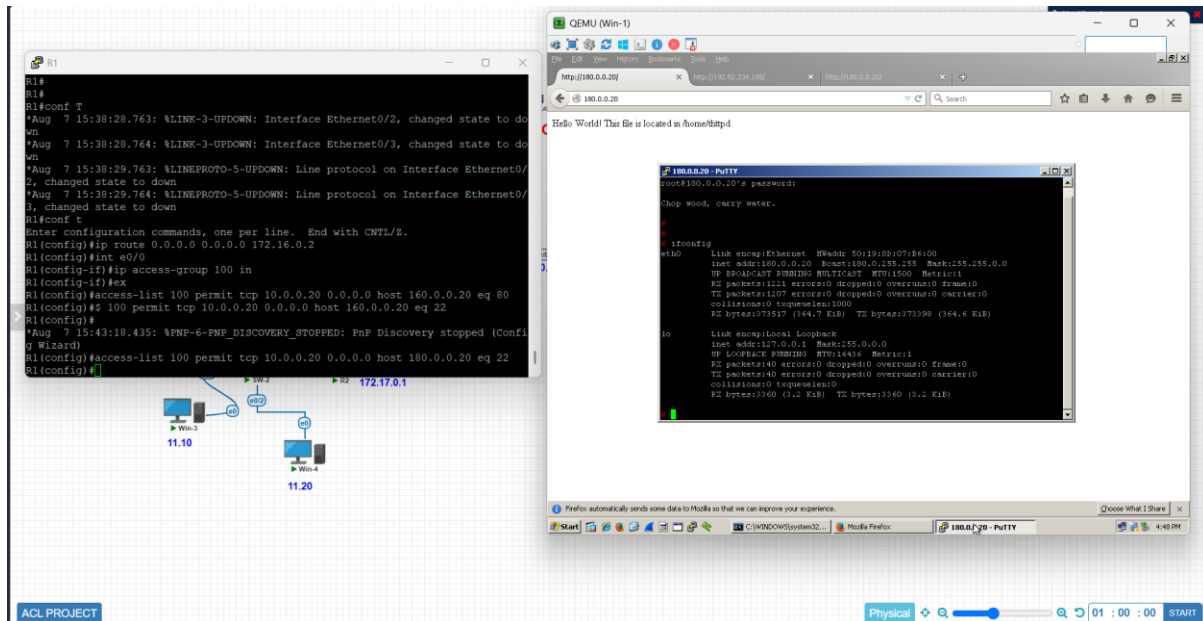


SCENERIO 3

Allow 10.0.0.20 to communicate facebook server using ssh & http

ACL CONFIGURATION IN ROUTER

PC OUTPUT - http & ssh

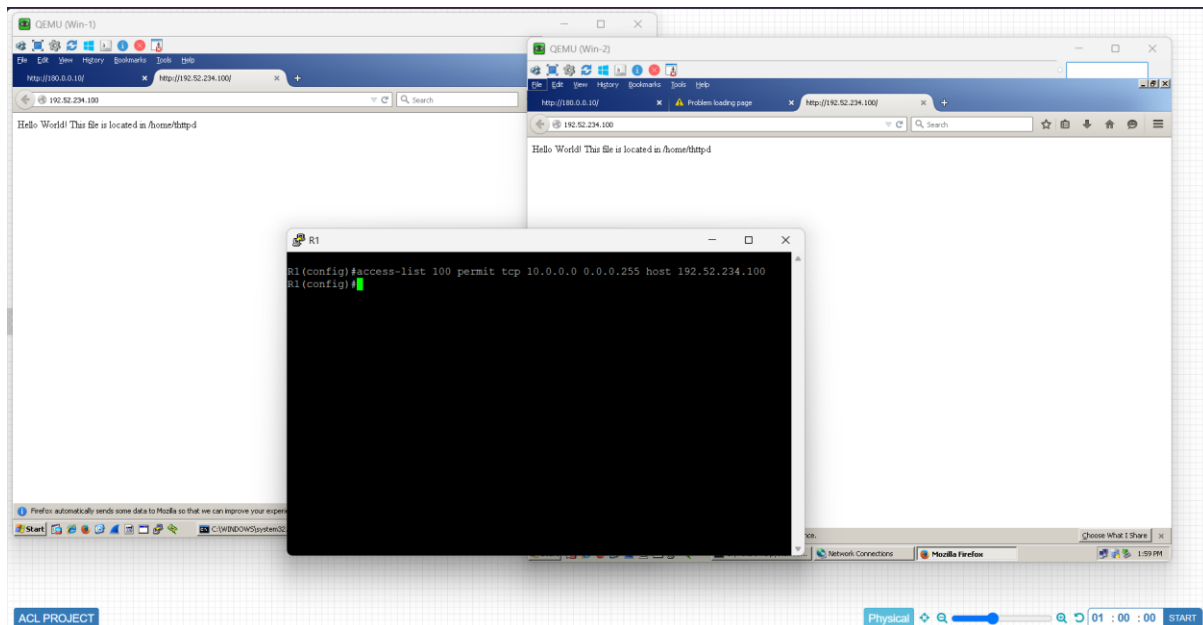


SCENERIO 4

Allow 10.0.0.0 to communicate snsite server using http

ACL CONFIGURATION IN ROUTER

PC OUTPUT - http

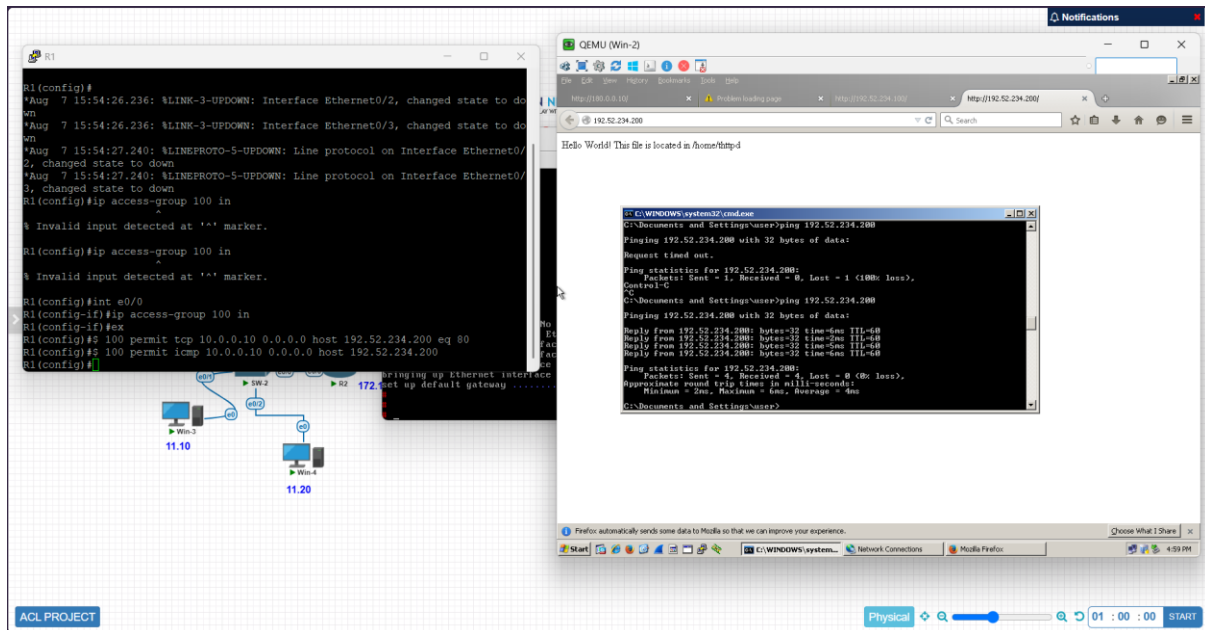


SCENERIO 5

Allow 10.0.0.10 to communicate snlabs server using http & icmp

ACL CONFIGURATION IN ROUTER

PC OUTPUT - http & icmp

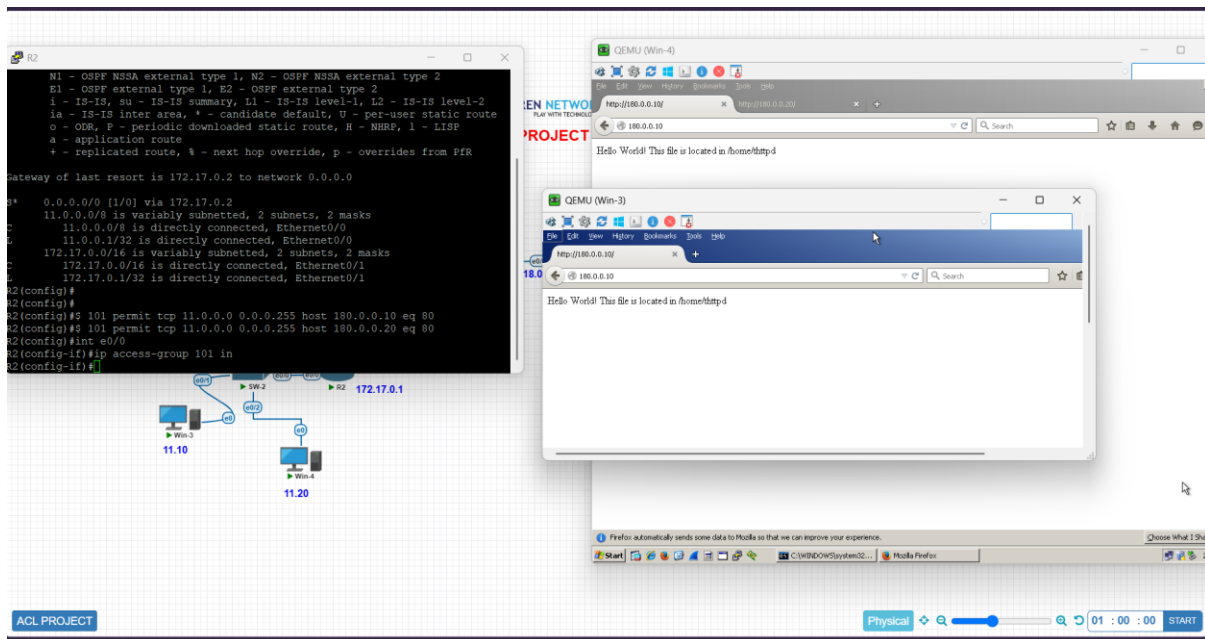


SCENARIO 6

Allow 11.0.0.0 to communicate instagram & facebook server using http

ACL CONFIGURATION IN ROUTER

PC OUTPUT - http → Instagram Server



ACL CONFIGURATION IN ROUTER

PC OUTPUT - http → Facebook Server

The screenshot displays a network simulation environment. On the left, a terminal window shows the configuration of Router R2. The configuration includes OSPF settings, interface definitions for Ethernet0/0 and Ethernet0/1, and an ACL named '101' that permits TCP traffic from the 11.0.0.0/8 network to the 180.0.0.0/8 network on port 80. The ACL is then applied to the Ethernet0/0 interface. A network diagram below the terminal shows two PCs, Win-3 (11.10) and Win-4 (11.20), connected to a switch, which is then connected to Router R2. On the right, two browser windows (QEMU Win-4 and QEMU Win-3) show the successful loading of 'Hello World' from a file located in /home/httpd, indicating that the HTTP traffic is permitted by the ACL.

SCENERIO 7

Allow 11.0.0.10 to ping facebook server

ACL CONFIGURATION IN ROUTER

RESPONSES FROM PC 3 & PC 4

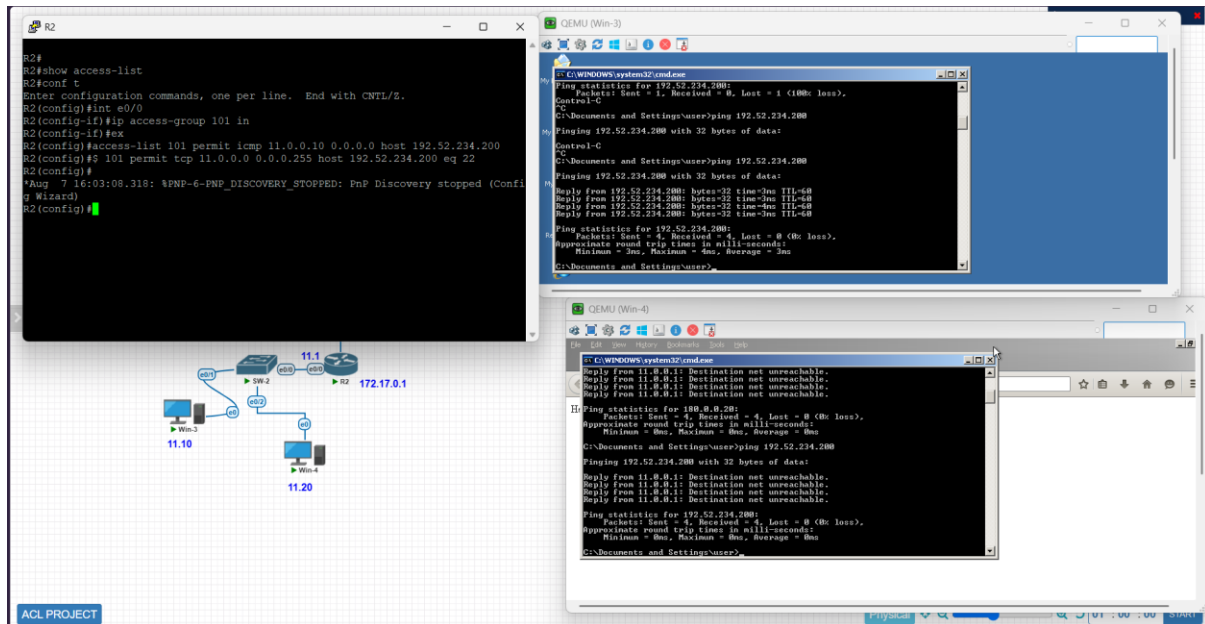
The screenshot displays a network simulation environment. On the left, a terminal window shows the configuration of Router R2. The configuration includes OSPF settings, interface definitions for Ethernet0/0 and Ethernet0/1, and an ACL named '101' that permits ICMP traffic from the 11.0.0.0/8 network to the 180.0.0.0/8 network. The ACL is then applied to the Ethernet0/0 interface. A network diagram below the terminal shows two PCs, Win-3 (11.10) and Win-4 (11.20), connected to a switch, which is then connected to Router R2. On the right, two command prompt windows (QEMU Win-4 and QEMU Win-3) show the results of a ping test to the IP address 180.0.0.20. Both PCs show successful ping results, indicating that the ICMP traffic is permitted by the ACL.

SCENARIO 8

Allow 11.0.0.10 to ping snlab server

ACL CONFIGURATION IN ROUTER

RESPONSES FROM PC 3 & PC 4

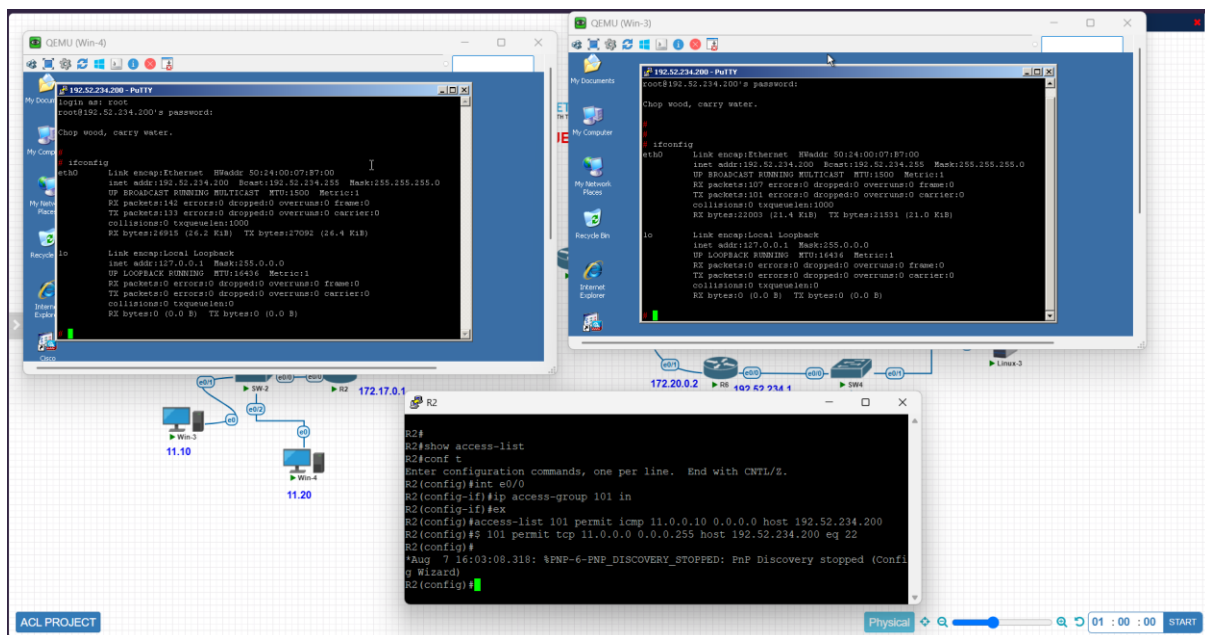


SCENARIO 9

Allow 11.0.0.0 to access snlab server

SSH ACCESS OF SNLAB SERVER IN PC 3

SSH ACCESS OF SNLAB SERVER IN PC 4



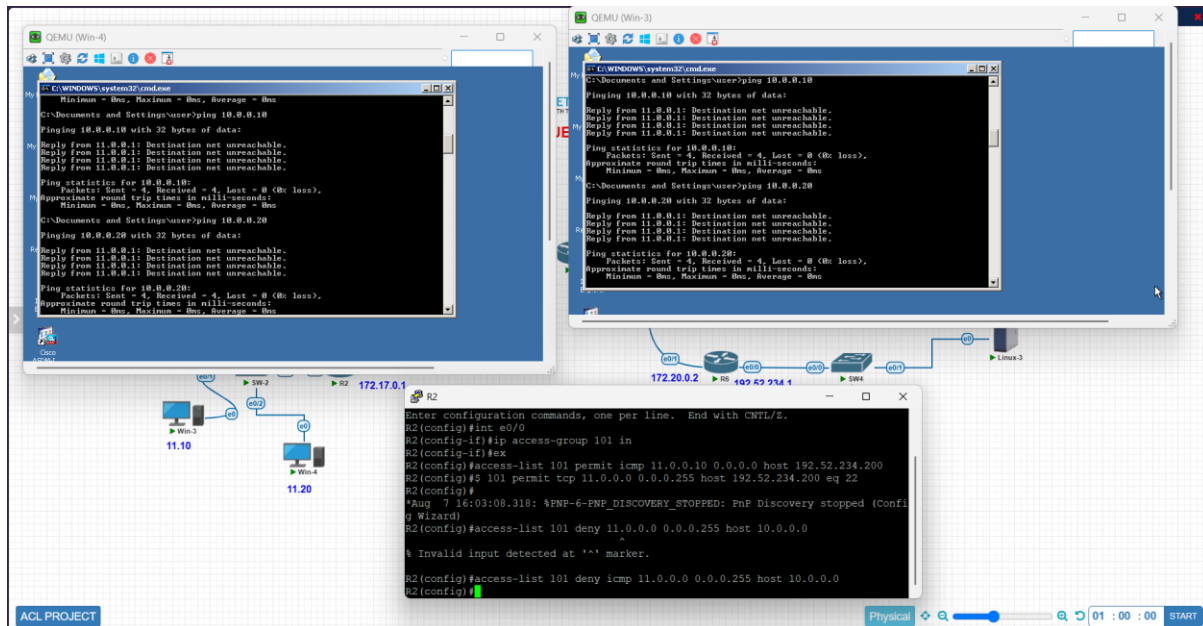
ACL CONFIGURATION IN ROUTER

SCENARIO 10

Deny 11.0.0.0 to access 10.0.0.0

RESPONSES FROM PC 3

RESPONSES FROM PC 4



ACL CONFIGURATION IN ROUTER

TASK 6: IDENTIFYING THE ERRORS

ERROR 1: Destination net unreachable.

- **Implicit Deny:** An ACL ends with an invisible rule that blocks all traffic not explicitly allowed.
- A router blocked the data packet using a filter rule.
- The router then generated an ICMP error message back to the computer to say the network is blocked or prohibited.

